



TIPPING BUCKET RAIN GAUGE

Complete Student Technical Documentation

Components · Assembly · Working Principle · Calibration · Data Acquisition

Type: Precision Tipping Bucket Rain Gauge

Accuracy: $\pm 1\%$ || Range: Up to 150
mm/hr || Collector Area: 200 cm²
Resolution: 0.2 mm or 0.5 mm per tip ||
Output: Reed Switch Pulse

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Chapter 1: Introduction — What Is a Rain Gauge?

A rain gauge is a scientific instrument used to measure the depth of liquid precipitation (rainfall) that falls over a specific area in a given time period. It is one of the most fundamental tools in meteorology — the science of weather and atmosphere.

Rainfall measurement matters across many fields of human activity:

- Agriculture: Farmers decide when and how much to irrigate based on how much rain already fell.
- Flood Control: Engineers design drainage systems using rainfall intensity data.
- Water Supply: Reservoirs and dams are managed using precipitation records.
- Weather Forecasting: Meteorologists feed rainfall data into prediction models.
- Climate Research: Long-term rainfall records reveal patterns and climate shifts.

❑ DID YOU KNOW?

Rain gauges have been used for over 2,000 years. Ancient India and Korea were among the first civilizations to measure rainfall systematically — to plan agriculture and taxation of farmland.

1.1 Types of Rain Gauges

Type	How It Works
Standard Cylinder Gauge	A simple graduated cylinder. Read manually. No automation.
Weighing Rain Gauge	Measures the weight of collected rainfall continuously.
Optical Rain Gauge	Laser beams detect falling drops. No moving parts.
Tipping Bucket Rain Gauge	See-saw mechanism counts measured tips. Automated. Most widely used globally.
Disdrometer	Measures raindrop size and velocity. Advanced research use only.

✔ THIS DOCUMENT

This documentation covers the Tipping Bucket Rain Gauge exclusively — the global standard for automated rainfall measurement in weather stations.

Chapter 2: Technical Specifications Explained

Every specification on the datasheet tells you something important about the instrument. Here is a complete explanation of each parameter:

Parameter	Value	What It Means
Instrument Type	Precision Tipping Bucket Rain Gauge	The measurement principle — a see-saw bucket that tips at a calibrated threshold.
Orifice Diameter	159.5 mm or 200 mm	The opening at the top. Wider = captures more rain.
Collector Area	200 cm ² or 314 cm ²	Cross-sectional area of the cylinder opening.
Sensing Technology	Magnetic reed switch	A magnetic-field-activated switch that generates a pulse each time the seesaw tips.
Measurement Range	Up to 150 mm/hour	Can accurately measure up to very heavy tropical storm rainfall intensities.
Resolution	0.2 mm or 0.5 mm per tip	The smallest measurable increment. Each tip of the seesaw = one resolution unit of rain.
Accuracy	±2% (traceable calibration)	Readings are within 2% of true value, verified against certified national standards.
Output Signal	Reed switch pulse count	Each tip sends a digital pulse to the data acquisition system.
Housing Material	UV-stabilized ABS polymer	Tough plastic that resists sunlight degradation and impact over years outdoors.
Color	Solar-reflective white	Reflects sunlight to prevent heating and evaporation of collected water.
Protection Rating	IP65	Fully dustproof and protected against water jets from any direction.

Chapter 3: Components

The tipping bucket rain gauge is made up of several carefully engineered components. Each serves a specific function. Below is a detailed explanation of every component used in the instrument.

3.1 Outer Cylinder (Rain Collector Housing)

Outer Cylinder

Dimensions: Height 295 mm | Inner Radius 81.75 mm | Collector Area: 200 cm² | Material: UV-stabilized ABS Polymer | Color: Solar-reflective white

The cylinder is the main outer body of the rain gauge — the tall tube you see from outside. Its top opening (the orifice) collects incoming rainfall. The inner walls funnel all collected water downward toward the seesaw mechanism. Made from ABS plastic with UV stabilizers to prevent cracking or yellowing from years of sunlight exposure. The white color reflects solar radiation so the gauge does not heat up and cause evaporation of collected water before it is measured.

3.2 Debriscap (Funnel Collector Cap)

Debriscap

Position: Top of the outer cylinder | Function: Primary rain collection and channeling

The Debriscap is the precision-shaped funnel cap that sits at the very top of the rain gauge. Its wide opening captures falling rain across the full 200 cm² collector area and channels it into a single, steady stream directed at the seesaw bucket below. The smooth internal geometry of the Debriscap is carefully designed to minimize water adhesion and surface tension effects — ensuring nearly all captured water drips cleanly into the measurement mechanism without sticking to the walls.

3.3 Bird Spike

Bird Spike

Material: Stainless steel or hard ABS | Position: Mounted on top center of the Debriscap | Function: Deterrent

The bird spike is a pointed spike (or array of small spikes) mounted at the center top of the gauge opening. Birds often perch on instruments mounted in open fields, and their droppings can block the funnel, contaminate the sensor, and cause false readings. The bird spike makes it uncomfortable for birds to land on the gauge without harming them. It is a simple but essential field accessory, especially in agricultural and natural environments where bird activity is high.

3.4 Mesh Net (Debris Filter)

Mesh Net

Material: Stainless steel wire mesh | Position: Inside the top funnel, below the Debriscap | Function: Debris filtration

The mesh net is a stainless steel wire screen fitted inside the funnel just below the Debriscap. It acts as the first line of defense for the measurement mechanism. The fine mesh allows water to pass through freely while blocking leaves, insects, twigs, dust clumps, and any other solid particles that could clog the funnel or jam the seesaw bucket. Stainless steel is used because it is corrosion-resistant — it will not rust even in constant contact with rainwater — and is easy to remove and clean during maintenance.

3.5 Seesaw (Tipping Bucket Mechanism)

Seesaw (Tipping Bucket)

Total Length: 119.31 mm | Width: 40.70 mm | Arm Span: 56.80 mm | Material: ABS Polymer | Magnet: Permanent magnet mounted on one arm

The seesaw is the heart of the entire instrument. It is a two-sided pivoting bucket — like a miniature children's see-saw — with a small funnel-shaped bucket on each end. Water drips into whichever bucket is currently raised. Once that bucket accumulates exactly enough water to represent 0.2 mm (or 0.5 mm) of rainfall across the 200 cm² collector area, its weight causes it to tip down. The other side rises and begins collecting. The tipped bucket drains through holes in its base and the cycle repeats. This mechanical action is the core measurement event.

❏ **VOLUME
PER TIP
CALCULATION**

Collector Area = 200 cm² | Resolution = 0.2 mm = 0.02 cm
Volume per tip = 200 cm² × 0.02 cm = 4 cm³ = 4 mL
For 0.5 mm resolution: Volume per tip = 200 × 0.05 = 10 mL

3.6 Clips (Seesaw Retention Clips)

Seesaw Retention Clips

Material: Stainless steel spring clips | Position: On the pivot axle of the seesaw | Function: Hold seesaw securely on its pivot axis

The retention clips are small stainless steel spring clips that hold the seesaw onto its pivot pin. Without these clips, the seesaw could vibrate off its axis under strong winds or during transport. The clips allow the seesaw to tip freely while keeping it firmly seated on the pivot. They are designed for tool-free removal — simply pinch and lift — so the seesaw can be taken out quickly during cleaning or calibration, then reinstalled securely.

3.7 Leveler (Spirit Level)

Spirit Level (Bubble Leveler)

Type: Circular bull's-eye spirit level | Position: Integrated into the base assembly | Function: Confirms the gauge is perfectly horizontal

The spirit level is a circular bubble level embedded in the base of the rain gauge. It contains a sealed liquid with a small air bubble. When the gauge is perfectly horizontal, the bubble sits exactly at the center of the circular target. If the gauge tilts in any direction, the bubble moves off-center, telling you which direction to adjust. This component is critical because even a small tilt

causes the seesaw to favor one side, making one bucket tip before it is fully filled — resulting in over-counting and inaccurate readings.

3.8 Long Adjustment Screws

Long Adjustment Screws

Material: Stainless steel | Count: 2 screws | Position: Mounted at the bottom of the seesaw, one under each bucket | Function: Control the amount of water each bucket holds before tipping

These two long screws are located underneath the seesaw — one below each bucket. By screwing them in or out, you raise or lower a small stop point that controls how far each bucket can tilt before it tips. This directly changes how much water the bucket holds before it reaches its tipping point. Screwing the adjuster in (upward) makes the bucket tip sooner — holding less water per tip. Screwing it out (downward) makes the bucket hold more water before tipping. This is the primary calibration tool: by adjusting these two screws, the technician sets the exact volume per tip to match the required resolution — 4 mL for 0.2 mm mode or 10 mL for 0.5 mm mode — achieving the specified $\pm 2\%$ accuracy.

3.9 Clamp (Base Mounting Clamp)

Base Mounting Clamp

Material: Galvanized steel or stainless steel | Function: Secures the rain gauge body to the mounting post or surface

The base clamp is the fastening component that locks the rain gauge firmly to its support post or mounting surface. It wraps around the lower body of the gauge and tightens via a bolt to grip the post. A secure clamp prevents the gauge from rotating (which would misalign the leveling) or being blown off by strong winds. The clamp must be tightened after leveling is complete so that the adjusted level position is locked in permanently. In field deployments, a loose clamp is one of the most common causes of measurement drift after initial installation.

3.10 Magnetic Reed Switch

Magnetic Reed Switch

Type: Normally-open (NO) reed switch | Signal: 1 pulse per tip | Housing: Glass-encapsulated sealed contacts | Output: 2-wire pulse to data acquisition

The reed switch is a tiny electronic switch enclosed in a sealed glass tube. Inside are two thin metal strips (reeds) that are normally apart (open circuit). The seesaw has a small permanent magnet attached to one arm. Every time the seesaw tips and the magnet swings past the reed switch, the magnetic field momentarily pulls the two reeds together, briefly closing the circuit and producing one electrical pulse. This pulse is the measurement event — it is sent to the Quectel data acquisition module via the output cable. The sealed glass housing makes it immune to moisture and corrosion.

Chapter 4: Working Principle

Now that you know every component, here is how they all work together to measure rainfall — from the first raindrop to the final data value.

4.1 Complete Measurement Cycle — Step by Step

- 1. Rain falls into the Debriscap opening (the wide top funnel of the gauge).
- 2. The mesh net filters out any debris — only clean water passes through.
- 3. Water drips down through the inner funnel onto the raised bucket of the seesaw.
- 4. As water accumulates, the bucket gets heavier. When it reaches exactly 4 mL (for 0.2 mm resolution), gravity tips that side down.
- 5. The other bucket swings up and starts collecting. The full bucket drains through its holes.
- 6. As the seesaw tips, the magnet sweeps past the reed switch, momentarily closing it — generating one electrical pulse.
- 7. The pulse travels through the output cable to the Quectel module's ADC input pin.
- 8. The Quectel module counts, timestamps, and transmits the data.
- 9. Total rainfall = Number of pulses × Resolution (0.2 mm or 0.5 mm per tip).

□ INTENSITY CALCULATION

If 50 tips are recorded in 1 hour (0.2 mm resolution): Total Rainfall = $50 \times 0.2 = 10$ mm
Intensity = $10 \text{ mm} \div 1 \text{ hr} = 10 \text{ mm/hr}$

4.2 Summary of Signal Flow

Stage	What Happens
1. Rain falls	Physical rainfall event → water enters Debriscap
2. Water accumulates	Seesaw bucket fills to calibrated threshold (4 mL or 10 mL)
3. Seesaw tips	Mechanical tipping action — magnet sweeps past reed switch
4. Reed switch closes	Momentary contact → 1 electrical pulse generated
5. Signal at ADC	Pulse arrives at Quectel module's ADC input pin
6. Count & transmit	Quectel counts pulses, calculates depth, transmits data

Chapter 5: Assembly

Assembly is the process of putting all the individual components together in the correct sequence to form a fully functional rain gauge. Each step must be completed in order — skipping or reversing steps can cause misalignment or incorrect calibration.

5.1 Assembly Sequence

10. **BASE PREPARATION:** Place the base assembly on a flat work surface. Insert the three long stainless steel leveling screws through the threaded base feet. Leave them at mid-travel so you have adjustment range in both directions.
11. **MOUNT CLAMP:** Fit the base mounting clamp around the base body. Do not fully tighten yet — leave it loose enough to allow leveling adjustments after the gauge is on its post.
12. **REED SWITCH INSTALLATION:** Press the magnetic reed switch into its mounting socket inside the lower housing at the factory-specified position. Route the 2-wire output cable through the cable gland at the base and out to the connector.
13. **SEESAW INSTALLATION:** Carefully lower the seesaw onto its pivot pin inside the lower housing. Ensure both sides pivot freely with no contact on the housing walls. The permanent magnet on the seesaw arm should be oriented to sweep within 1–2 mm of the reed switch face as it tips.
14. **RETENTION CLIPS:** Snap the stainless steel retention clips onto both ends of the pivot pin. Confirm the seesaw cannot be lifted off the pivot but still tips smoothly and freely.
15. **MESH NET:** Drop the stainless steel mesh net into the top of the funnel housing. It should sit flat and cover the full inner diameter without gaps at the edges.
16. **DEBRISCAP:** Press the Debriscap funnel cap onto the top of the cylinder housing. It should click or lock securely with no gaps that would allow rain to bypass the mesh.
17. **BIRD SPIKE:** Insert or screw the bird spike into the center post on top of the Debriscap. Ensure it is vertical and firmly seated.
18. **OUTER CYLINDER:** Slide the outer cylinder over the internal assembly and secure to the base with the locking ring or screws provided.
19. **LEVELING:** Mount the gauge on its post. Use the spirit level bubble and the long leveling screws to achieve perfect horizontal alignment. Tighten the mounting clamp securely once level is confirmed.
20. **CABLE CONNECTION:** Connect the output cable to the Quectel module's ADC input pin and ground pin. Verify polarity.
21. **FUNCTIONAL TEST:** Pour water slowly into the top of the assembled gauge. Confirm tips occur and each tip generates a pulse detectable at the Quectel module.

5.2 Critical Assembly Checks

Check Item	Required Condition
Seesaw pivot friction	Seesaw must swing freely under its own weight with no sticking
Magnet-to-reed distance	Within 1–2 mm at closest point during tip — verify with test tip

Clip security	Clips must prevent seesaw from lifting off the pivot under vibration
Mesh net flat seating	No gaps around the edge — all water must pass through mesh
Spirit level centered	Bubble must be centered after mounting clamp is fully tightened
Cable gland sealed	Tighten cable gland to prevent water ingress along the cable
Bird spike vertical	Must not lean — a leaning spike may cause rain to run off asymmetrically

Chapter 6: Calibration

Calibration is the process of verifying and adjusting the rain gauge so that each seesaw tip corresponds exactly to the specified rainfall depth — 0.2 mm or 0.5 mm over the 200 cm² collector area. A gauge that is not properly calibrated gives misleading data regardless of how well it is built.

6.1 Why Calibration Is Critical

The seesaw tips when the water in the bucket reaches a specific mass. This mass depends on:

- The exact position of the seesaw's center of gravity (balance point)
- The volume of each bucket cavity
- The friction of the pivot bearing
- Environmental factors like temperature (which slightly changes water density)

Even small manufacturing variations between units mean each gauge must be individually calibrated. The specified accuracy is $\pm 2\%$, meaning a gauge measuring a true 100 mm storm should read between 98 mm and 102 mm.

6.2 Calibration Tool — The Burette

□ WHAT IS A BURETTE?

A burette is a long, graduated glass tube with a stopcock (tap) at the bottom, used in laboratories to dispense precise, measured volumes of liquid. It is the standard tool for rain gauge calibration because it can deliver exact volumes of water — drop by drop if needed — with an accuracy of ± 0.05 mL.

For calibration of this rain gauge, the burette is mounted vertically on a stand directly above the gauge funnel. Water is released from the burette at a slow, controlled rate to simulate low-intensity rainfall. The volume delivered per tip is computed and compared to the specification.

6.3 Calibration Procedure for 0.2 mm Resolution

At 0.2 mm resolution, each tip must occur after exactly 4 mL of water has entered the gauge ($200 \text{ cm}^2 \times 0.02 \text{ cm} = 4 \text{ mL}$).

22. Fill the burette with clean distilled water. Note the starting volume reading.
23. Position the burette directly above the gauge funnel opening. Ensure the gauge is level.
24. Open the stopcock slowly to allow a gentle, steady drip — simulating light rainfall.
25. Count the number of tips produced as water drips in. Count minimum 20 tips for statistical accuracy.
26. Note the burette reading after 20 tips. Volume used = Start reading – End reading.
27. Calculate volume per tip: Volume per tip = Total volume ÷ Number of tips
28. Expected result: 4.00 mL per tip (acceptable range: 3.92 mL to 4.08 mL for $\pm 2\%$ tolerance).
29. If volume per tip is outside range, adjust the seesaw balance point using the calibration adjustment screw until the correct threshold is achieved.
30. Repeat the test to confirm the adjusted value is within specification.

✓
**ACCEPTANCE
CRITERION**
(0.2 mm)

Volume per tip must be between 3.92 mL and 4.08 mL. If 20 tips use between 78.4 mL and 81.6 mL of water from the burette — the gauge PASSES.

6.4 Calibration Procedure for 0.5 mm Resolution

At 0.5 mm resolution, each tip must occur after exactly 10 mL of water has entered the gauge ($200 \text{ cm}^2 \times 0.05 \text{ cm} = 10 \text{ mL}$).

31. Fill the burette with clean distilled water. Note the starting volume reading.
32. Position the burette above the gauge funnel. Ensure the gauge is level.
33. Open the stopcock slowly for a steady drip rate.
34. Count 20 tips.
35. Calculate volume per tip = Total volume used ÷ 20 tips.
36. Expected result: 10.00 mL per tip (acceptable range: 9.80 mL to 10.20 mL for $\pm 2\%$ tolerance).
37. If outside range, adjust seesaw balance and retest until within specification.
38. Record calibration date, result, and technician name on the calibration certificate.

✓
**ACCEPTANCE
CRITERION**
(0.5 mm)

Volume per tip must be between 9.80 mL and 10.20 mL. If 20 tips use between 196.0 mL and 204.0 mL from the burette — the gauge PASSES.

6.5 Comparison: 0.2 mm vs 0.5 mm Calibration

Parameter	0.2 mm Mode	0.5 mm Mode
Resolution	0.2 mm per tip	0.5 mm per tip
Volume per tip	4.00 mL	10.00 mL

TIPPING BUCKET RAIN GAUGE

Accepted range ($\pm 2\%$)	3.92 – 4.08 mL	9.80 – 10.20 mL
Burette volume for 20 tips	~80 mL	~200 mL
Best use case	Research, precise monitoring	General agriculture, hydrology
Pulse rate at 150 mm/hr	~1 pulse / 5.76 sec	~1 pulse / 14.4 sec